Experiment -2

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Design a UI where users recall visual elements (e.g., icons or text chunks). Evaluate the effect of chunking on user memory.

FRAME 1:
INSTRUCTION PAGE:

Purpose of the Screen:

- Introduces the task clearly with the title “MEMORY TASK.”
- Provides simple navigation options: Play and Help.
- Creates curiosity and encourages users to begin the task.

Key UI Elements:

- Title (“MEMORY TASK”) placed at the top for instant recognition.
- Play Button – Primary action, highlighted to guide users to start.
- Help Button – Secondary action for instructions or assistance
- Visual Theme (Cosmetics & Soft Colors) – Creates a friendly and engaging atmosphere.

UX Benefits:

- Minimal content reduces cognitive load.
- Clear call-to-action improves usability.
- Attractive visuals increase user engagement

FRAME 2:

CHUNKING PHASE:

Analysis of the Memory Recall Task - Chunking Phase Screen

This screen represents the **Chunking Phase** of a **Memory Recall Task**, where users observe and memorize different icons within a limited time. Below is a breakdown of its key components:

Clear and Sequential Numbering:

- Instructions are broken into six numbered steps.
- Each step communicates a single action, improving clarity

Logical Grouping of Information:

- Observation Phase (Steps 1–2): Users understand what they will see
- Memorization Phase (Step 3): Encourages focusing on remembering items.
- Recall Phase (Steps 4–5): Explains how recall will happen.
- Performance Metric (Step 6): Emphasizes accuracy over speed.

Visual Hierarchy:

- Bold heading “INSTRUCTIONS” attracts attention.
- Large readable text ensures clarity..
- Play button placed at the bottom signals progression..

UX & Cognitive Benefits:

- Reduces confusion before starting the task.
- Helps users mentally prepare for memory recall.
- Improves comprehension through chunked instructions.

FRAME 3:

RECALL PHASE:

Analysis of the Memory Recall Task - Selection Phase

This image displays the **Recall Phase** results of a cognitive experiment, specifically focusing on how grouping information affects memory accuracy. This particular test was conducted under the **"Grouped by Category"** condition.

Purpose of the screen

- Represents the memory retrieval stage of the experiment.
- Users select the icons they remember seeing in the chunking phase.
- Evaluates short-term memory retention and the effectiveness of chunking.

Key elements and UI components

Top-level metadata and navigation

- Header label “Recall Phase” clearly indicates the current stage.
- Instructional text guides users to select remembered icons.
- The layout keeps focus on the selection task without distractions.

Primary metric scorecards

- Uses a traffic-light color system:
  - Green: Correct selections
  - Red: Incorrect selections
  - Yellow/Orange: Missed selections

Results grid

- Each tile shows a checkmark or cross indicating correctness.
- Color-coded borders indicate:
  - Green: Correctly identified icons
  - Red: False positives
  - Dimmed: Unselected or incorrect items

Icon selection grid

- Displays a structured grid containing:
  - Icons shown in the study phase
  - Additional distractor icons
- All icons are presented uniformly without indicating correctness.
- Users select icons based purely on memory recall.

Design Language & UX

- Dark Mode Background: Uses a deep charcoal/navy to reduce eye strain and make the colored borders pop.
- High-Affordance Feedback: The use of both color (red/green) and symbols (check/X) ensures the interface is accessible for users with color vision deficiencies.
- Symmetry: The grid is perfectly aligned, allowing the user to scan their "mistakes" vs. "successes" systematically.

How the selection phase works

- Users recall icons seen during the chunking phase.
- They select the icons they remember from the grid.
- Distractor icons test the accuracy of recall.
- Clicking the “Submit” button finalizes the user’s responses.

Cognitive and UX benefits

- Tests genuine memory recall without visual hints.
- Encourages careful decision-making based on memory.
- Simple layout minimizes cognitive load.
- Gamified interaction keeps users engage

FRAME 4:

RESULT PAGE:

Analysis of the Memory Recall Task - Score & Feedback Screen

This screen represents the **Score & Feedback Phase** of the **Memory Recall Task**, where users receive their performance evaluation based on the selections made in the previous **Recall Phase**.

Purpose of the screen

- Displays the overall memory performance of the user.
- Shows how many items were correctly recalled out of the total.
- Allows users to retry the test or proceed further.

Performance breakdown

- Shows the total number of correct answers.
- Displays accuracy as a fraction or percentage (e.g., 3/8).
- Does not visually mark individual icons as right or wrong.
- Focuses on overall performance rather than detailed error analysis.

Cognitive and UX benefits

- Score-based feedback provides clear performance insight.
- Avoids confusion by not highlighting individual mistakes.
- Encourages users to improve by retaking the test.
- Clean and minimal design keeps feedback easy to understand.
- Supports fair comparison between chunked and non-chunked conditions.

PROTOTYPE LINK:
https://www.figma.com/proto/rmsZ8jCa28bX86lrNdlEFR/Untitled?page-id=0%3A1&node-id=21-2&p=f&viewport=-133%2C-974%2C0.58&t=89JVcPfClRxNGATL-1&scaling=scale-down&content-scaling=fixed&starting-point-node-id=21%3A2